

IT3091 – Bank Marketing

EDA Insight Log

This log records each notable finding from exploratory data analysis, the evidence behind it, and its implication for preprocessing or modeling.

#	Insight	Evidence	Implication for Preprocessing / Modeling
1	Dataset is a clean, complete table with no null cells at load time, but not free of data-quality issues.	45,211 rows x 17 columns (bank-full.csv). df.isnull().sum() shows 0 for all columns as loaded, but categorical fields use a separate "unknown" label / are recoded to NaN by ucimlrepo (see #2).	Do not assume "no NaNs" means "no missing data" — the unknown category must be checked and handled explicitly.
2	Several categorical fields contain missing/unknown values, at very different scales.	job: 288 missing (0.64%); education: 1,857 (4.11%); contact: 13,020 (28.80%); poutcome: 36,959 (81.75%).	job/education: low impact, impute mode or keep as own category. contact: too large a share to drop or impute confidently, keep as an explicit "unknown" category. poutcome: extremely high missingness, but not random - see row 7 - treat as its own "no_prior_campaign" category rather than imputing.
3	Severe target class imbalance.	y = no: 39,922 rows (88.3%); y = yes: 5,289 rows (11.7%).	Use a stratified train/test split (and stratified k-fold for model comparison); evaluate with precision, recall, F1, and PR-AUC rather than accuracy alone; consider class weighting in model comparison.
4	Numeric features show many IQR-flagged outliers, but at very different rates and for different reasons.	age: 487 (1.08%); balance: 4,729 (10.46%); campaign: 3,064 (6.78%); pdays: 8,257 (18.26%); previous: 8,257 (18.26%).	age/balance/campaign: outliers appear genuine (older clients, high/negative balances, persistently-called clients) — prefer capping or robust scaling for linear/distance-based models; leave as-is for tree-based models.

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			pdays/previous: the flagged "outliers" are not anomalies at all — see row 7.
5	pdays contains a large sentinel spike at -1.	Histogram of pdays shows a dominant mass at -1, representing clients never previously contacted, rather than a true "days since contact" value.	Treat -1 as a categorical flag (e.g. create a binary "previously_contacted" feature) rather than feeding pdays as-is into a numeric/distance-based model.
6	duration is a strong but unusable predictor — confirmed leakage.	Median call duration: 164 seconds for y = no vs 426 seconds for y = yes (~2.6x longer for subscribers).	Exclude duration from the deployable model, since it is only known after a call takes place. May be reported separately as a diagnostic-only upper-bound benchmark, with this limitation stated explicitly.
7	Discovery: the pdays/previous "outliers" and the poutcome missingness are the same subpopulation.	pdays and previous each have exactly 8,257 non-sentinel values (18.26%); poutcome is present for 100% - 81.75% = 18.25% of rows - matching almost exactly. Confirmed with <code>df[df.pdays==-1].poutcome.isnull().mean() = ~1.0</code> and <code>df[df.pdays!=-1].poutcome.isnull().mean() = ~0.0</code> .	poutcome is missing precisely when a client was never contacted in a previous campaign - it is structurally missing, not random. Engineer one binary "previously_contacted" feature from pdays/previous, and treat poutcome's missing values as a meaningful "no_prior_campaign" category. Do not apply IQR outlier removal to pdays/previous - the flagged rows are the informative previously-contacted minority, not errors.
8	poutcome = "success" is one of the strongest, genuinely non-leaky predictors of subscription.	Countplot (Cell 8) shows clients whose previous campaign outcome was "success" subscribe at a markedly higher rate than clients with "failure", "other", or no prior campaign.	Retain and encode poutcome carefully, including the new "no_prior_campaign" category from row 7. This is a high-value, ethically uncomplicated feature, unlike duration — worth highlighting in the model comparison and recommendation stages.

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9	previous and pdays are moderately correlated; no other numeric pair is.	Pearson correlation heatmap (Cell 9): previous vs pdays $r = 0.45$ — the only numeric pair above ~ 0.4 . All other pairs (age, balance, campaign) show weak correlation.	Moderate, not severe, multicollinearity - no need to drop either feature outright, but this reinforces the row 7 decision to engineer a single "previously_contacted" binary flag rather than feeding both raw, sentinel-heavy columns into distance-based models.